VOLT-FORGE

A visual workspace for battery trade-offs

CASE STUDY 2

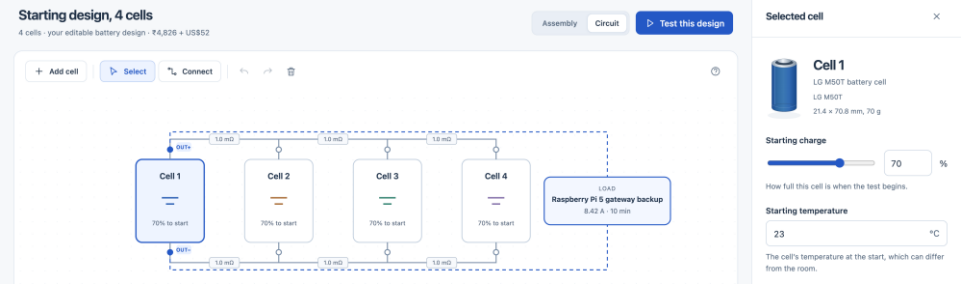
Two cells are a simulation candidate

Smallest qualifying count tested (1-4 cells): half the baseline's cell cost and mass, with less electrical margin.

Pi 5 backup | LG M50T | 25 W supply-rating scenario | 10 min | 23°C | 70% starting SOC

90% converter assumed | 3.3 V study floor | 8.42 A constant-current proxy | actual Pi draw not measured

BUILD, TEST, UNDERSTAND



Original V4 editor; illustrative circuit, not the study table.

For students and lab instructors: edit a circuit, inspect current, voltage and simulated temperature, then compare alternatives and export the evidence.

BASELINE + TWO ALTERNATIVES

Parallel design	Required passes	Worst cell current	Worst cell temperature	Cell mass	Listed cell cost
4 cells	51/51	3.34 A	26.1°C	280 g	₹4,824
3 cells	42/42	3.86 A	27.7°C	210 g	₹3,618
2 cells	**33/33**	**5.49 A**	**31.8°C**	**140 g**	**₹2,412**

Frozen V4: same-end outputs; mismatch, thermal and converter sensitivity cases. Counts vary by cell position. Temperatures simulated; cell-only cost and mass.

EVIDENCE

15/15 numerical checks passed

All 12 comparison cases missed voltage and temperature bands. Measured accuracy is not established.

LIMITS THAT MATTER

Conditional, not build-ready

No spare cell. Board current limits unverified. Virtual cutoffs modelled; physical protection has not been demonstrated.

USER + BUYER HYPOTHESIS

Start with teaching labs

Free student use; paid lab licences untested. No observed user trials yet. Next user task: complete and explain a comparison.

NEXT Measure load and converter behavior; bench-check with supervised cutoffs; observe student tasks.

Source + code: github.com/jaikaushik-prog/VOLT-FORGE